

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is the principal reason why it is important to study how the Kalman filter is derived?

1 / 1 point

- ☐ Because it is a good opportunity to practice and refine math skills.
- ☐ Because it is valuable to see how the Kalman filter solved the problem of optimal state estimation, for historic perspective and interest.
- ☐ Because Kalman filters are not actually used for state estimation and must be modified so that they can be used for other applications.
- ☒ Because many applications violate the assumptions made in the derivation and require that we are able to make modifications to the derivation.

✔ **Correct**
Yes. This is correct.

2. What are examples of applications that violate the assumptions made when deriving the standard Kalman filter? (Select all that apply.)

1 / 1 point

- ☒ State-estimation applications for nonlinear systems.

✔ **Correct**
Yes, this is one application that violates the assumptions made when deriving the standard Kalman filter. (You will study this topic in the third course.)

- ☐ State-estimation applications for linear models.

- ☒ Parameter-estimation applications.

✔ **Correct**
Yes, this is one application that violates the assumptions made when deriving the standard Kalman filter. (It turns out to be a nonlinear problem that you will study in the third course.)

- ☒ Target-tracking applications where measurements arrive at the Kalman filter out of sequence.

✔ **Correct**
Yes, this is one application that violates the assumptions made when deriving the standard Kalman filter.

3. Which application of Kalman filters seeks to estimate the location of the platform that implements and executes the Kalman filter using local sensors?

1 / 1 point

- ☒ Navigation.
- ☐ State estimation for nonlinear systems.
- ☐ Parameter estimation.
- ☐ Target tracking.

✔ **Correct**
Yes. This is correct.